Impact of 6G Network Performance on Manufacturing Efficiency in Smart Factories

Executive Summary — Evidence-Based Insights for Smart Manufacturing Network Monitoring | Prepared for government stakeholders by Isha Kalra, October 2026

100,000	50	69 days	−0.0011	−0.0024
observations	machines	1 Jan–10 Mar 2025	Spearman correlation: latency vs. production speed	Spearman correlation: packet loss vs. error rate

PURPOSE

This project examines whether network delay (latency) and data loss (packet loss) are associated with manufacturing efficiency, production speed, operating errors, and product defects in a smart-factory dataset. It aims to identify patterns that can improve monitoring and guide further investigation.

WHY THIS MATTERS

Connected factories rely on timely, reliable communication between equipment and monitoring systems, so network conditions are one of several factors worth checking when performance changes. Machine condition and operating mode also matter. Knowing what the evidence does and does not show supports better diagnosis without assuming the network is the cause of production problems.

DATA AND APPROACH

The analysis covers 100,000 observations from 50 machines (1 January–10 March 2025): network latency and packet loss, efficiency status (Low, Medium, High), production speed, error and defect rates, operating mode (Active, Idle, Maintenance), and temperature, vibration, and power use. Methods were descriptive and diagnostic. A Network Stability Index, a dataset-relative score combining latency and packet loss with equal weights (higher = more stable), split observations into three network-quality groups. Results were also compared by operating mode and within comparable (not identical) temperature, vibration, and power profiles. A Spearman rank correlation, from −1 to +1, shows whether two measures rise and fall together consistently; a value near 0 means no consistent relationship.

KEY FINDINGS

- Network profile: latency averaged 25.6 ms (range 1–50 ms) and packet loss 2.49% (range 0–5%). Daily stability scores stayed close to 0.50, with no distinct stable or unstable periods.
- Latency and production speed: the Spearman correlation was −0.0011 (essentially none). Across the full 1–50 ms range, the fitted change is only about 0.5 units per hour (under 0.2% of average output); this is a descriptive association, not an expected production loss. No sustained latency range linked to slower production was found.
- Packet loss and errors: the Spearman correlation was −0.0024 (essentially none); error rates stayed between 7.47% and 7.56% across all packet-loss ranges.
- Defects: defect rates stayed between 4.95% and 5.07% across packet-loss ranges, including 5.01% in the highest range (4.5–5%).
- Efficiency: Low efficiency accounts for 77.83% of observations, Medium 19.19%, and High only 2.99% (rounded; true shares total 100%). These shares were virtually identical across network-quality groups (Low: 77.75%–77.89%) and within comparable mechanical-condition profiles (other factors were not controlled). High efficiency is rare, so compare against this baseline.

IMPLICATIONS

Network performance is worth monitoring, but this dataset does not establish it as an independent cause of manufacturing efficiency, speed, errors, or defects. Network measurements should be read alongside machine condition, operating mode, production behavior, and quality outcomes, rather than used alone to diagnose operational problems. No latency-tolerance benchmark or packet-loss risk zone could be established; this does not show that none exists in real factories.

RECOMMENDATIONS

- Integrated monitoring: track network, production, quality, and machine-condition indicators together.
- Trend-based investigation: investigate persistent changes and deviations rather than adopting fixed thresholds derived from this dataset.
- Context-aware diagnosis: consider operating mode and mechanical conditions when investigating changes in output or quality.
- Better data: collect network, machine, production, and defect measurements with synchronized timestamps and documented provenance.
- Validate before setting rules: test any proposed threshold on additional operational data and relevant engineering standards before adoption.
- Continued evaluation: use future data to check whether patterns hold across machines, conditions, and longer periods.

LIMITATIONS

The data are observational, so no cause-and-effect conclusions can be drawn. Network-quality groups and the stability index are relative to this dataset, not engineering standards or 6G thresholds. The data have no real-time/delayed label, cover limited ranges (latency up to 50 ms, packet loss up to 5%), and contain a repeated block of 2,880 records on 1 March 2025 that was kept and flagged. Many values are evenly spread across fixed ranges, which is typical of simulated data, and the dataset's origin is undocumented. The analysis does not independently verify that the data were collected from a real 6G deployment.

CONCLUSION

The analysis found no strong relationship between the examined network measures and manufacturing outcomes, but it shows the value of integrated, context-aware monitoring. Additional synchronized data and further validation are recommended before operational thresholds are set or causal claims are made.